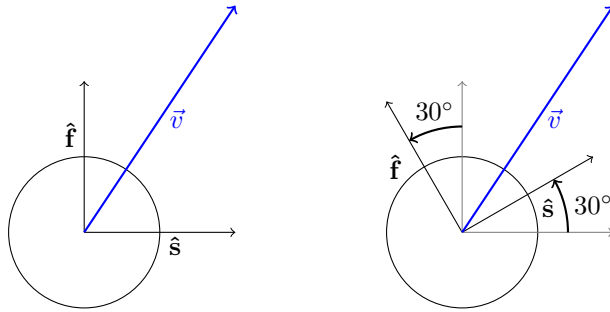


1 Determining velocity in reference frame of vessel

Vector $\vec{v}$ is our velocity.

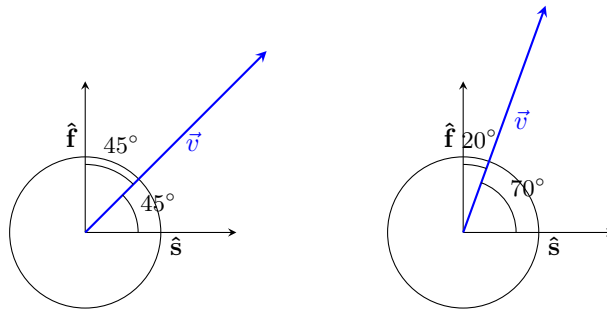
Vectors $\hat{\mathbf{f}}$ and $\hat{\mathbf{s}}$ are forward and starboard directions of a vessel. So $\hat{\mathbf{f}}$ would be the 'nose'.



In the reference frame of the ground (or planet) $\vec{v}$ does not change direction as the vessel rotates. Notice in the drawing $\hat{\mathbf{f}}$ and $\hat{\mathbf{s}}$ rotate 30 degrees. But $\vec{v}$ remains static. (Imagine the illustrations as a top down view from an external observer.)

As $\hat{\mathbf{f}}$ and $\hat{\mathbf{s}}$ are unit vectors, they will have a magnitude of 1. While $\vec{v}$ will have a variable magnitude representing groundspeed in m/s. The goal is to calculate $\vec{v}$ but represented in $\hat{\mathbf{f}}$ and $\hat{\mathbf{s}}$.

A dot product tells us how much two vectors point in the same direction. Imagine walking along $\vec{v}$ at a 45° , you will walk equally along the vectors $\hat{\mathbf{f}}$ and $\hat{\mathbf{s}}$. If we change the angle of $\vec{v}$ to 20° from $\hat{\mathbf{f}}$, then we will walk further along $\hat{\mathbf{f}}$ than $\hat{\mathbf{s}}$.



Because of this relation, we can simply do as following:

Foward speed

$$\vec{f} = \hat{\mathbf{f}} \cdot \vec{v}$$

Starboard speed

$$\vec{s} = \hat{\mathbf{s}} \cdot \vec{v}$$

Because of how dot products work, if $\vec{v}$ points opposite of $\hat{\mathbf{f}}$ and $\hat{\mathbf{s}}$, the resulting dot product will be negative. This is good, as we can use this output to represent the vessel as going backwards or left (starboard representing right).

2 Translating relative velocity to counteractive control inputs